UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE • DEPARTMENT OF COMPUTER SCIENCE

User Manual: Liquor Store System
Operating Guide for Owners and Employees — Academic Version 1.0

1. SYSTEM OVERVIEW & PURPOSE

The **Liquor Store System** is a software solution engineered to streamline the commercial and administrative workflows of a traditional liquor store. Designed to run locally via an interactive and secure console interface, it empowers owners and employees to centralize control over inventory, suppliers, cash and credit sales, customer payments, daily cash reconciliation, and financial indicators.

2. PREREQUISITES & EXECUTION

To operate the system, you must have the **Java Development Kit (JDK 17 or higher)** and the **NetBeans** IDE installed. To launch the application, open the project in NetBeans, ensure the file encoding is set to UTF-8 (to properly display Spanish console characters), and execute the main class located within the ec.edu.espe.systemliquorstore.view package.

3. MAIN MENU NAVIGATION GUIDE

Upon starting the system, the main menu is displayed with options numbered 0 through 10. Enter the number corresponding to your desired operation. You may type /cancelar at any input prompt to abort an ongoing operation and safely return to the main menu.

OPTION	MENU / MODULE	FUNCTIONAL DESCRIPTION
1	Products and inventory	Product registration, listing, pricing/stock updates, and manual entry/exit stock movements.
2	Low stock alerts	Automatic query identifying products that have reached or fallen below their minimum stock threshold.
3	Suppliers and orders	Management of suppliers, creation of purchase orders, and receiving inventory deliveries.
4	Register sale	Complete execution of the sales cycle (cash or credit, discount application, and age verification).

5	Customers and payments	Customer registration, consultation of outstanding debt balances, and payment history tracking.

MAIN MENU NAVIGATION (CONTINUED)

OPTION	MENU / MODULE	FUNCTIONAL DESCRIPTION
6	Daily cash	Cash drawer opening, recording cash expenses, physical counting, and daily closing differences.
7	Gross profit and margin	Financial calculation of sales profitability and global gross margin over a specified date range.
8	Consult sales by seller	Filtering and breakdown of historical transactions executed specifically by the owner or employee.
9	Best-selling products	Descending ranking of product units sold within a selected time frame.
10	Load demonstration data	Quickly populates the system with a fictional initial scenario (demo products, supplier, and customer).
0	Exit	Safely terminates the current work session after requiring user confirmation.

4. DETAILED SALES WORKFLOW (OPTION 4)

Registering a sale is the core process of the business. It strictly follows a defined sequence to guarantee inventory integrity, accurate accounting, and regulatory compliance:

1. **Seller Identification:** Select the responsible party: 1. Propietario (Owner) or 2. Empleado (Employee).
2. **Date & Payment Method:** Enter the date in YYYY-MM-DD format (press Enter to default to today) and choose between Contado (Cash) or Crédito (Credit). A credit sale requires selecting a previously registered customer.
3. **Adding Products:** Enter the product code and the desired quantity. The system validates sufficient stock availability in real-time. Press Enter without typing a code to finish adding items.
4. **Authorized Discount:** Optionally, apply a discount percentage authorized by the owner (a numeric value between 0 and less than 100).

5. **Age Verification:** Input the customer's birth date. The system calculates the exact age and strictly requires the customer to be 18 years or older to proceed.
6. **Final Confirmation:** Review the detailed on-screen summary (subtotal, discount, total amount, seller, and payment type) and confirm the transaction by typing s (sí/yes).

5. DAILY CASH MANAGEMENT & CLOSING (OPTION 6)

The daily cash module allows for strict control of the liquor store's cash flow through the following key operations:

- **Opening Cash (Apertura de Caja):** Must be registered at the start of the workday, specifying the date and the initial cash balance inside the drawer.
- **Cash Expenses (Gastos en Efectivo):** Allows the recording of minor cash outflows for daily operational needs, requiring a brief description and the exact amount spent.
- **Physical Count & Closing:** At the end of the shift, the cashier inputs the total amount physically counted in bills and coins. The system calculates the *Expected Cash* using the formula:

  > Expected Cash = Opening Cash + Cash Sales + Debt Payments − Cash Expenses

- **Difference:** The system displays the exact result indicating if the drawer is Balanced (Cuadrada), has a Surplus (Sobrante), or has a Shortage (Faltante).

6. BEST PRACTICES & CONSOLE HANDLING

To ensure smooth operations and prevent data entry errors during store hours, adhere to the following operational recommendations:

Numeric & Decimal Formats: Always enter monetary amounts using a dot . as the decimal separator (e.g., 25.50). Never use commas or currency symbols like "$".

Canceling Operations: If you make a mistake or need to interrupt a data entry process, simply type the command /cancelar at any prompt to safely return to the main menu without altering records.

Data Persistence: All successful operations (product updates, sales, payments) are automatically saved to local JSON files. Always close the program using option 0 to guarantee data is synchronized and safely stored.

Critical Inventory Management: Periodically review Option 2 in the main menu to proactively restock products before they fall below their minimum operational threshold.